

SQL DATA ANALYSIS PROJECT

HR Employee Attrition Analysis

*25 Business Questions Solved Using
MySQL*

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PROJECT OVERVIEW



1. PROJECT OBJECTIVE

Analyze employee attrition data using MySQL and solve 25 business questions to identify patterns and key factors influencing employee turnover.



2. DATASET OVERVIEW



1,470
Total Employees



237
Employees Left



16.12%
Overall Attrition Rate



34
Employee Attributes
(Columns)



3. ANALYSIS AREAS



Department Analysis



Job Role Analysis



Salary Analysis



Demographic Analysis
(Age, Gender)



Overtime & Business
Travel Analysis



Job Satisfaction &
Marital Status



4. SQL SKILLS USED



SELECT



WHERE



GROUP BY



HAVING



ORDER BY



CASE



JOINS



SUBQUERIES



CTEs



WINDOW
FUNCTIONS



Tools Used: MySQL | Environment: MySQL Workbench

Retrieve the total number of employees

```
SELECT  
    COUNT(*) AS total_emp  
FROM  
    hr_attrition;
```

Result Grid	
	total_emp
▶	1470

Calculate the total number of employees who left the organization

```
SELECT  
    COUNT(*) AS left_emp  
FROM  
    hr_attrition  
WHERE  
    attrition = 'yes';
```

Result Grid		
	left_emp	
▶	237	

Calculate the overall employee attrition rate

```
SELECT  
  (SUM(CASE  
    WHEN attrition = 'yes' THEN 1  
    ELSE 0  
  END) / COUNT(EmployeeNumber)) * 100 AS attrition_rate  
FROM  
  hr_attrition;
```

Result Grid	
	attrition_rate
▶	16.1224

Identify the department with the highest number of employees

```
SELECT
    department, COUNT(*) AS total_emp
FROM
    hr_attrition
GROUP BY department
ORDER BY total_emp DESC
LIMIT 1;
```

Result Grid		
Filter Rows:		
	department	total_emp
▶	Research & Development	961

Identify the job role with the highest number of employees

```
SELECT  
    jobrole, COUNT(*) AS total_emp  
FROM  
    hr_attrition  
GROUP BY JobRole  
ORDER BY total_emp DESC  
LIMIT 1;
```

Result Grid		 Filter Rows:
	jobrole	total_emp
▶	Sales Executive	326

Analyze the total number of employees and attrition count for each department

```
SELECT
    department,
    SUM(CASE
        WHEN attrition = 'yes' THEN 1
        ELSE 0
    END) AS dept_attrition_count
FROM
    hr_attrition
GROUP BY department;
```

Result Grid			Filter Rows:
	department	dept_attrition_count	
▶	Sales	92	
	Research & Development	133	
	Human Resources	12	

Calculate the attrition rate for each department

```
SELECT
    department,
    (SUM(CASE
        WHEN attrition = 'yes' THEN 1
        ELSE 0
    END) / COUNT(*)) * 100 AS dept_attrition_rate
FROM
    hr_attrition
GROUP BY department;
```

Result Grid			Filter Rows:	
	department	dept_attrition_rate		
▶	Sales	20.6278		
	Research & Development	13.8398		
	Human Resources	19.0476		

Calculate the attrition rate for each job role

```
SELECT
    jobrole,
    (SUM(CASE
        WHEN attrition = 'yes' THEN 1
        ELSE 0
    END) / COUNT(*)) * 100 AS attrition_by_role
FROM
    hr_attrition
GROUP BY JobRole
ORDER BY attrition_by_role DESC;
```

Result Grid			Filter Rows:
	jobrole	attrition_by_role	
▶	Sales Representative	39.7590	
	Laboratory Technician	23.9382	
	Human Resources	23.0769	
	Sales Executive	17.4847	
	Research Scientist	16.0959	
	Manufacturing Director	6.8966	
	Healthcare Representative	6.8702	
	Manager	4.9020	
	Research Director	2.5000	

Analyze the relationship between overtime and employee attrition

```
-- Attrition_Count
SELECT
    overtime,
    SUM(CASE
        WHEN attrition = 'yes' THEN 1
        ELSE 0
    END) AS attrition_count
FROM
    hr_attrition
GROUP BY OverTime;
```

	overtime	attrition_count
►	Yes	127
	No	110

```
-- Attrition_Rate
SELECT
    overtime,
    SUM(CASE
        WHEN attrition = 'yes' THEN 1
        ELSE 0
    END) / COUNT(*) * 100 AS attrition_rate
FROM
    hr_attrition
GROUP BY OverTime;
```

	overtime	attrition_rate
►	Yes	30.5288
	No	10.4364

Analyze the average monthly income of employees based on their attrition status

```
SELECT
    attrition, AVG(monthlyincome) AS avg_salary
FROM
    hr_attrition
GROUP BY attrition;
```

Result Grid			Filter Row
	attrition	avg_salary	
▶	Yes	4787.0928	
	No	6832.7397	

Analyze the distribution of employees and attrition across different age groups

```
SELECT
CASE
    WHEN age BETWEEN 18 AND 25 THEN '18-25'
    WHEN age BETWEEN 26 AND 35 THEN '26-35'
    WHEN age BETWEEN 36 AND 45 THEN '36-45'
    WHEN age BETWEEN 46 AND 55 THEN '46-55'
    ELSE '56+'
END AS age_group,
COUNT(*) AS total_employees,
SUM(CASE
    WHEN attrition = 'Yes' THEN 1
    ELSE 0
END) AS attrition_count
FROM
    hr_attrition
GROUP BY age_group
ORDER BY age_group;
```

Result Grid			
Filter Rows:			
	age_group	total_employees	attrition_count
▶	18-25	123	44
	26-35	606	116
	36-45	468	43
	46-55	226	26
	56+	47	8

Analyze the relationship between job satisfaction and employee attrition

```
SELECT
    jobsatisfaction,
    COUNT(*) AS total_employee,
    SUM(CASE
        WHEN attrition = 'yes' THEN 1
        ELSE 0
    END) AS attrition_count
FROM
    hr_attrition
GROUP BY jobsatisfaction
ORDER BY jobsatisfaction;
```

Result Grid			
Filter Rows:			
	jobsatisfaction	total_employee	attrition_count
▶	1	289	66
	2	280	46
	3	442	73
	4	459	52

Analyze the relationship between business travel and employee attrition

```
SELECT
    businesstravel,
    COUNT(*) AS total_employee,
    SUM(CASE
        WHEN attrition = 'yes' THEN 1
        ELSE 0
    END) AS total_attrition
FROM
    hr_attrition
GROUP BY businesstravel
ORDER BY total_attrition desc;
```

Result Grid			
Filter Rows:			
	businesstravel	total_employee	total_attrition
▶	Travel_Rarely	1043	156
	Travel_Frequently	277	69
	Non-Travel	150	12

Analyze the attrition rate based on marital status

```
SELECT
    maritalstatus,
    COUNT(*) AS total_employee,
    SUM(CASE
        WHEN attrition = 'yes' THEN 1
        ELSE 0
    END) AS total_attrition,
    (SUM(CASE
        WHEN attrition = 'yes' THEN 1
        ELSE 0
    END) / COUNT(*)) * 100 AS attrition_rate
FROM hr_attrition
GROUP BY maritalstatus;
```

Result Grid				
		Filter Rows:	Export:	
	maritalstatus	total_employee	total_attrition	attrition_rate
▶	Single	470	120	25.5319
	Married	673	84	12.4814
	Divorced	327	33	10.0917

Analyze the attrition rate based on gender

```
SELECT
    gender,
    COUNT(*) AS total_employee,
    SUM(CASE
        WHEN attrition = 'yes' THEN 1
        ELSE 0
    END) AS attrition_count,
    (SUM(CASE
        WHEN attrition = 'yes' THEN 1
        ELSE 0
    END) / COUNT(*)) * 100 AS attrition_rate
FROM
    hr_attrition
GROUP BY gender;
```

Result Grid					Filter Rows:	Export:
	gender	total_employee	attrition_count	attrition_rate		
▶	Female	588	87	14.7959		
	Male	882	150	17.0068		

Determine the top 3 job roles with the highest attrition rate

```
SELECT
    jobrole,
    COUNT(*) AS total_emp,
    (SUM(CASE
        WHEN attrition = 'yes' THEN 1
        ELSE 0
    END) / COUNT(*)) * 100.0 AS attrition_rate
FROM
    hr_attrition
GROUP BY JobRole
ORDER BY attrition_rate DESC
LIMIT 3;
```

Result Grid			
Filter Rows:			
	jobrole	total_emp	attrition_rate
►	Sales Representative	83	39.75904
	Laboratory Technician	259	23.93822
	Human Resources	52	23.07692

Determine the department with the highest attrition rate using a ranking function

```
WITH t1 AS (  
    SELECT  
        department,  
        COUNT(*) AS total_employee,  
        SUM(CASE  
            WHEN attrition = 'yes' THEN 1  
            ELSE 0  
        END) AS attrition_count,  
        (SUM(CASE  
            WHEN attrition = 'yes' THEN 1  
            ELSE 0  
        END) / COUNT(*)) * 100 AS attrition_rate  
    FROM hr_attrition  
    GROUP BY department  
,  
t2 AS (  
    SELECT *,  
        RANK() OVER (ORDER BY attrition_rate DESC) AS rank_no  
    FROM t1  
)  
SELECT *  
FROM t2  
WHERE rank_no = 1;
```

Result Grid


Filter Rows:

Export:


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	department	total_employee	attrition_count	attrition_rate	rank_no
▶	Sales	446	92	20.6278	1

Analyze the relationship between overtime and job satisfaction to identify high-risk employee groups

```
SELECT
    overtime,
    jobsatisfaction,
    COUNT(*) AS total_employee,
    SUM(CASE
        WHEN attrition = 'yes' THEN 1
        ELSE 0
    END) AS attrition_count,
    (SUM(CASE
        WHEN attrition = 'yes' THEN 1
        ELSE 0
    END) / COUNT(*)) * 100 AS attrition_rate
FROM
    hr_attrition
GROUP BY OverTime , JobSatisfaction
ORDER BY attrition_rate DESC;
```

Result Grid					
Filter Rows:					
Export:  Wrap Cell Content: ☐					
	overtime	jobsatisfaction	total_employee	attrition_count	attrition_rate
►	Yes	2	69	26	37.6812
	Yes	1	84	30	35.7143
	Yes	3	121	41	33.8843
	Yes	4	142	30	21.1268
	No	1	205	36	17.5610

Calculate the percentage contribution of each department to the total number of employees who left

```
SELECT
    Department,
    SUM(CASE
        WHEN Attrition = 'Yes' THEN 1
        ELSE 0
    END) AS attrition_count,

    (SUM(CASE
        WHEN Attrition = 'Yes' THEN 1
        ELSE 0
    END) /
    (SELECT COUNT(*)
     FROM hr_attrition
     WHERE Attrition = 'Yes')) * 100 AS contribution_percentage

FROM hr_attrition
GROUP BY Department
ORDER BY contribution_percentage DESC;
```

Result Grid				Filter Rows:	Export:	Wra
	Department	attrition_count	contribution_percentage			
▶	Research & Development	133	56.1181			
	Sales	92	38.8186			
	Human Resources	12	5.0633			

Rank job roles within each department based on their attrition rate

```
WITH role_attrition AS (  
    SELECT  
        department,  
        JobRole,  
        COUNT(*) AS total_employee,  
        SUM(CASE  
            WHEN attrition = 'yes' THEN 1  
            ELSE 0  
        END) AS attrition_count,  
        (SUM(CASE  
            WHEN attrition = 'yes' THEN 1  
            ELSE 0  
        END) / COUNT(*)) * 100 AS attrition_rate  
    FROM hr_attrition  
    GROUP BY department, JobRole  
)  
SELECT  
    department,  
    JobRole,  
    total_employee,  
    attrition_count,  
    attrition_rate,  
    DENSE_RANK() OVER (  
        PARTITION BY department  
        ORDER BY attrition_rate DESC  
    ) AS d_rank  
FROM role_attrition;
```

	department	JobRole	total_employee	attrition_count	attrition_rate	d_rank
▶	Human Resources	Human Resources	52	12	23.0769	1
	Human Resources	Manager	11	0	0.0000	2
	Research & Development	Laboratory Technician	259	62	23.9382	1
	Research & Development	Research Scientist	292	47	16.0959	2
	Research & Development	Manufacturing Director	145	10	6.8966	3

Compare average salary, average age, and average years at company between employees who stayed and employees who left

```
SELECT
  CASE
    WHEN attrition = 'yes' THEN 'Left'
    ELSE 'Stayed'
  END AS employee_status,

  ROUND(AVG(monthlyincome), 2) AS avg_salary,
  ROUND(AVG(age), 2) AS avg_age,
  ROUND(AVG(yearsatcompany), 2) AS avg_years_at_company

FROM hr_attrition

GROUP BY
  CASE
    WHEN attrition = 'yes' THEN 'Left'
    ELSE 'Stayed'
  END;
```

Result Grid					Filter Rows:	Export:	Wrap C
	employee_status	avg_salary	avg_age	avg_years_at_company			
▶	Left	4787.09	33.61	5.13			
	Stayed	6832.74	37.56	7.37			

Identify employee groups with high attrition based on multiple factors such as overtime, job satisfaction, age, and job role

```
SELECT
    overtime,
    jobsatisfaction,

    CASE
        WHEN age < 25 THEN '18-24'
        WHEN age BETWEEN 25 AND 34 THEN '25-34'
        WHEN age BETWEEN 35 AND 44 THEN '35-44'
        ELSE '45+'
    END AS age_group,

    jobrole,

    COUNT(*) AS total_employee,
```

```
        SUM(
            CASE
                WHEN attrition = 'yes' THEN 1
                ELSE 0
            END
        ) AS attrition_count,

    ROUND(
        SUM(
            CASE
                WHEN attrition = 'yes' THEN 1
                ELSE 0
            END
        ) / COUNT(*) * 100,
        2
```

```
    ) AS attrition_rate

FROM hr_attrition

GROUP BY
    overtime,
    jobsatisfaction,
    age_group,
    jobrole

ORDER BY attrition_rate DESC;
```

Result Grid

Filter Rows:

Export:

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Fetch rows:

	overtime	jobsatisfaction	age_group	jobrole	total_employee	attrition_count	attrition_rate
	Yes	3	18-24	Sales Representative	1	1	100.00
	Yes	1	35-44	Human Resources	1	1	100.00
	Yes	1	25-34	Human Resources	2	2	100.00
	Yes	3	25-34	Human Resources	1	1	100.00
▶	Yes	2	35-44	Sales Representative	1	1	100.00

Result Grid

Filter Rows:

Export:

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Fetch rows:

	overtime	jobsatisfaction	age_group	jobrole	total_employee	attrition_count	attrition_rate
	Yes	1	18-24	Laboratory Technician	3	3	100.00
	Yes	3	18-24	Laboratory Technician	2	2	100.00
	Yes	2	18-24	Sales Representative	4	4	100.00
	Yes	3	18-24	Manufacturing Director	1	1	100.00
▶	Yes	3	25-34	Manager	1	1	100.00

Use a CTE to calculate and compare department-wise attrition rates

```
WITH department_attrition AS (  
    SELECT  
        department,  
        COUNT(*) AS total_employee,  
        SUM(  
            CASE  
                WHEN attrition = 'yes' THEN 1  
                ELSE 0  
            END  
        ) AS attrition_count  
    FROM hr_attrition  
    GROUP BY department  
)
```

```
SELECT  
    department,  
    total_employee,  
    attrition_count,  
    ROUND(  
        (attrition_count / total_employee) * 100,  
        2  
    ) AS attrition_rate  
FROM department_attrition  
ORDER BY attrition_rate DESC;
```

Result Grid



Filter Rows:

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	department	total_employee	attrition_count	attrition_rate
►	Sales	446	92	20.63
	Human Resources	63	12	19.05
	Research & Development	961	133	13.84

Use window functions to rank departments and job roles based on attrition

```
WITH department_attrition AS (  
  SELECT  
    department,  
    COUNT(*) AS total_employee,  
    SUM(  
      CASE  
        WHEN attrition = 'yes' THEN 1  
        ELSE 0  
      END  
    ) AS attrition_count,  
    ROUND(  
      SUM(CASE WHEN attrition = 'yes' THEN 1 ELSE 0 END)  
      / COUNT(*) * 100,  
      2  
    ) AS attrition_rate  
  FROM hr_attrition  
  GROUP BY department
```

```
),  
jobrole_attrition AS (  
  SELECT  
    jobrole,  
    COUNT(*) AS total_employee,  
    SUM(  
      CASE  
        WHEN attrition = 'yes' THEN 1  
        ELSE 0  
      END  
    ) AS attrition_count,  
    ROUND(  
      SUM(CASE WHEN attrition = 'yes' THEN 1 ELSE 0 END)  
      / COUNT(*) * 100,  
      2  
    ) AS attrition_rate
```

```
FROM hr_attrition  
GROUP BY jobrole  
)  
  
SELECT  
  department,  
  total_employee,  
  attrition_count,  
  attrition_rate,  
  DENSE_RANK() OVER (  
    ORDER BY attrition_rate DESC  
  ) AS department_rank  
FROM department_attrition  
  
ORDER BY department_rank;
```

Result Grid



Filter Rows:

Export:



Wrap Cell Content:



	department	total_employee	attrition_count	attrition_rate	department_rank
►	Sales	446	92	20.63	1
	Human Resources	63	12	19.05	2
	Research & Development	961	133	13.84	3